An Open-Source Quality Management System

1. Executive Summary

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2. The Problem

2.1 The Reproducibility Crisis

Academia is struggling with a reproducibility crisis. Published estimates of irreproducibility range from 51% (Hartshorne and Schachner 2012) to 89% (Begley and Ellis 2012) and it is estimated that irreproducible preclinical research wastes approximately 28 billion dollars a year in the US alone (Freedman, Cockburn, and Simcoe 2015).

Many of the systemic causes of irreproducibility can ultimately be traced to an underlying lack of a standards and best practices framework (Freedman, Cockburn, and Simcoe 2015; Institute 2013) or to poor data management (Soundararajan and Mishra 2023). Additionally, while academia has peer review for *results*, it rarely peer reviews *process*.

2.2 The Translation Gap

Furthermore, translating validated and reproducible results into the clinic is a daunting and difficult task, known as "the valley of death" (Coller and Califf 2009).

Academic institutions lack the experience and infrastructure needed to produce regulated products, or evidence supporting them, in a regulatory-compliant manner (Coller and Califf 2009; Stamler, Taber, and Califf 2003).

Data are assessed for fitness for use and reliability before they can inform a regulatory decision, and protocols and analysis plans are expected to be finalized before analyses are conducted (U.S. Food and Drug Administration 2023). Consequently, data originating in academia which have not been generated under prospectively defined and documented procedures cannot serve as the primary basis for a determination of substantial evidence (United States 2026). At best they can provide corroborative support, and only where they are "carefully conducted and documented". Reports lacking the details that permit scientific evaluation are not considered at all (United States 2026).

2.3 The Shared Root Cause

3. The Solution

4. Why *Open Source*

4.1 Eliminating Duplicated Effort

4.2 Best Practices Become Reviewable and Contestable

4.3 Rationale Is Preserved, Not Just the Requirement

4.4 Network Effects

4.5 A Teaching Artifact

4.6 Extensibility Across Domains

5. Prior Work

6. Implementation: QMS as Code

6.1 Minimal Primitives

6.2 Deliberate Vendor Neutrality

6.3 Cost

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